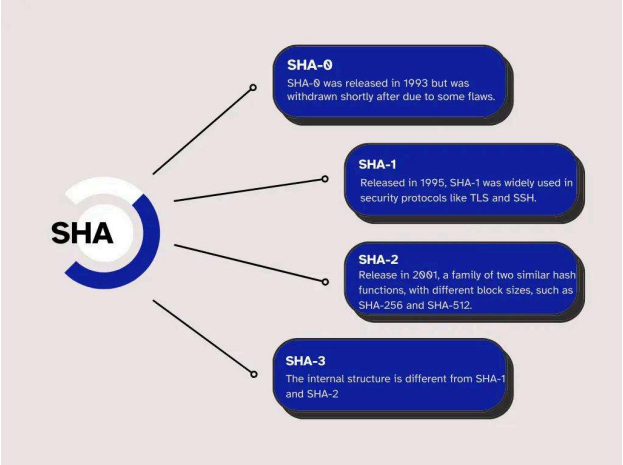
2. Historical Background and Applications of SHA

2.1 History of SHA



The Secure Hash Algorithm (SHA) family was developed by the National Institute of Standards and Technology (NIST) to provide a secure way of converting data into fixed-length hashes for integrity and security purposes.

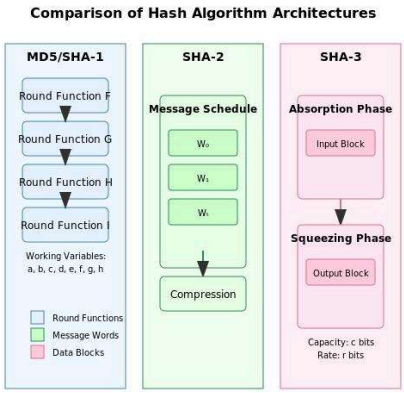
The first version, **SHA-0**, was introduced in 1993. However, it was soon found to have design flaws, so it was withdrawn and replaced by **SHA-1** in 1995. SHA-1 became extremely popular and was widely used in digital certificates, software integrity checks, and version control systems.

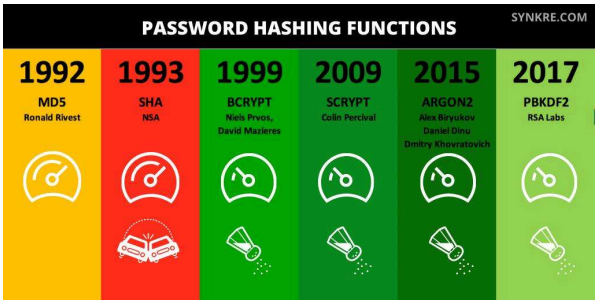
Over time, advances in computational power made SHA-1 vulnerable to collision attacks. A major milestone was the **SHAttered attack (2017)**, where researchers demonstrated a practical collision. After this, SHA-1 was officially deprecated for secure applications.

To address these weaknesses, NIST introduced the **SHA-2 family in 2001**, which includes SHA-224, SHA-256, SHA-384, and SHA-512. These algorithms provided much stronger security and are still widely used today.

Later, in 2015, **SHA-3** was introduced based on the Keccak algorithm. Unlike SHA-1 and SHA-2, it uses a completely different structure (sponge construction). SHA-3 acts as a backup standard in case vulnerabilities are found in SHA-2, ensuring long-term cryptographic security.

Overall, the SHA family has evolved step by step to handle increasing security threats and computational capabilities.





2.2 Applications of SHA

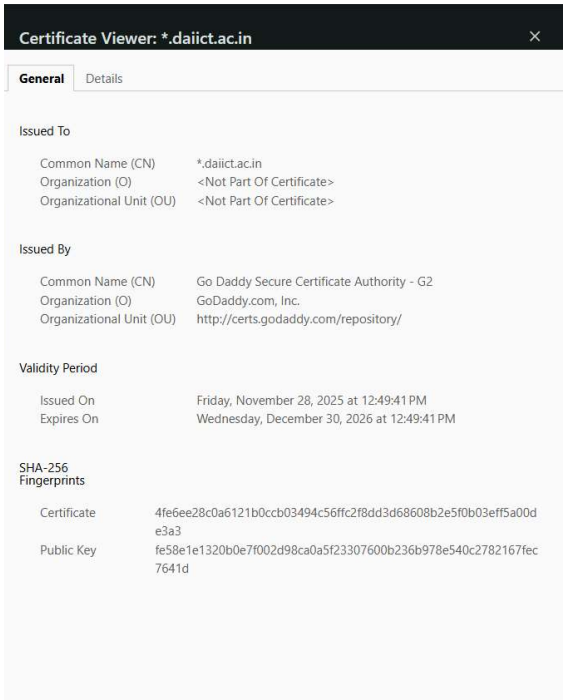
SHA algorithms are used in many real-world systems where data security and integrity are important.

1. Secure Communication (HTTPS / SSL Certificates)
Modern web security relies heavily on SHA-2 (especially SHA-256).

- Used in HTTPS connections to ensure data is not tampered with
- Replaced SHA-1 in browser certificates

Examples:

- Google Chrome, Mozilla Firefox (certificate validation)



Certificate Viewer: *.daiict.ac.in

General | Details

Issued To

Common Name (CN)	*.daiict.ac.in
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	Go Daddy Secure Certificate Authority - G2
Organization (O)	GoDaddy.com, Inc.
Organizational Unit (OU)	http://certs.godaddy.com/repository/

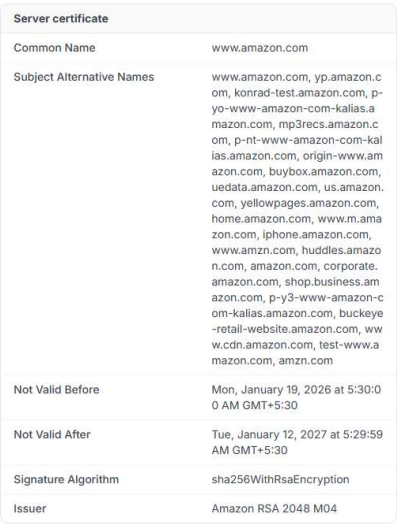
Validity Period

Issued On	Friday, November 28, 2025 at 12:49:41 PM
Expires On	Wednesday, December 30, 2026 at 12:49:41 PM

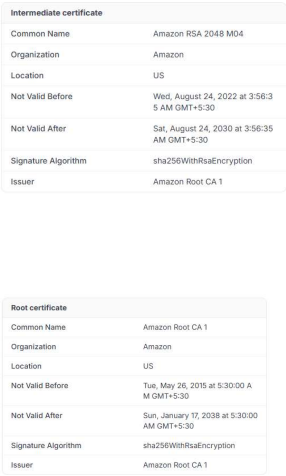
SHA-256 Fingerprints

Certificate	4fe6ee28c0a6121b0ccb03494c56ffc2f8dd3d68608b2e5f0b03eff5a00de3a3
Public Key	fe58e1e1320b0e7f002d98ca0a5f23307600b236b978e540c2782167fec7641d

- Websites using HTTPS (banks, e-commerce platforms)



Server certificate	
Common Name	www.amazon.com
Subject Alternative Names	www.amazon.com, yp.amazon.com, konrad-test.amazon.com, p-yo-www-amazon-com-kalias.amazon.com, mp3recs.amazon.com, p-nt-www-amazon-com-kalias.amazon.com, origin-www.amazon.com, buybox.amazon.com, uedata.amazon.com, us.amazon.com, yellowpages.amazon.com, home.amazon.com, www.m.amazon.com, iphone.amazon.com, www.amzn.com, huddles.amazon.com, amazon.com, corporate.amazon.com, shop.business.amazon.com, p-y3-www-amazon-com-kalias.amazon.com, buckeye-retail-website.amazon.com, www.cdn.amazon.com, test-www.amazon.com, amzn.com
Not Valid Before	Mon, January 19, 2026 at 5:30:00 AM GMT+5:30
Not Valid After	Tue, January 12, 2027 at 5:29:59 AM GMT+5:30
Signature Algorithm	sha256WithRsaEncryption
Issuer	Amazon RSA 2048 M04



Intermediate certificate	
Common Name	Amazon RSA 2048 M04
Organization	Amazon
Location	US
Not Valid Before	Wed, August 24, 2022 at 3:56:35 AM GMT+5:30
Not Valid After	Sat, August 24, 2030 at 3:56:35 AM GMT+5:30
Signature Algorithm	sha256WithRsaEncryption
Issuer	Amazon Root CA 1

Root certificate	
Common Name	Amazon Root CA 1
Organization	Amazon
Location	US
Not Valid Before	Tue, May 26, 2015 at 5:30:00 AM GMT+5:30
Not Valid After	Sun, January 17, 2038 at 5:30:00 AM GMT+5:30
Signature Algorithm	sha256WithRsaEncryption
Issuer	Amazon Root CA 1

https://seositecheckup.com/seo-audit/ssl-checker-and-https-test/amazon.com

---

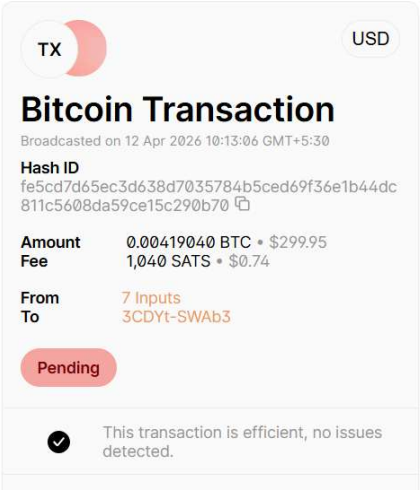
2. Blockchain and Cryptocurrencies

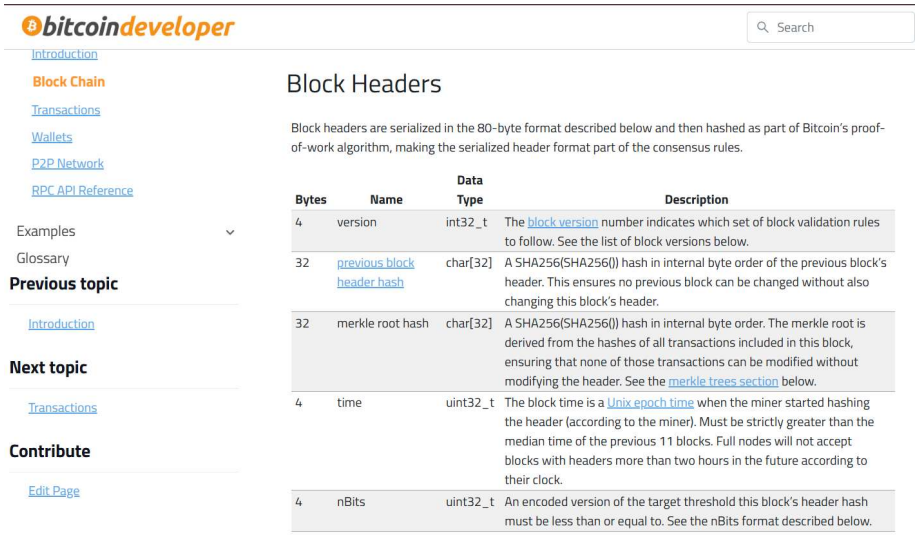
SHA-256 is a core component in blockchain systems.

- Used for mining and block hashing
- Ensures immutability of transactions

Examples:

- Bitcoin (uses SHA-256 for proof-of-work)
- Many blockchain systems and wallets





https://developer.bitcoin.org/reference/block_chain.html

---

3. Password Storage (with enhancements)
SHA is used along with salting and other techniques to store passwords securely.

- Prevents storing plain-text passwords
- Often combined with stronger schemes like bcrypt or Argon2

Examples:

- Backend authentication systems in websites
- Linux password hashing (older systems used SHA-based hashing)

```
user:$6$salt$hashedpassword...
```

---

4. File Integrity and Software Distribution
SHA is used to verify that files have not been altered.

- Hash values are published alongside downloads
- Users can verify integrity after download

Examples:

- Linux ISO downloads (SHA-256 checksums)
- Git (uses SHA-1, transitioning to SHA-256)

```
PS C:\Users\HP\Downloads> certutil -hashfile ubuntu-22.04.5-desktop-amd64.iso SHA256
SHA256 hash of ubuntu-22.04.5-desktop-amd64.iso:
bfd1cee02bc4f35db939e69b934ba49a39a378797ce9aee20f6e3e3e728fefbf
CertUtil: -hashfile command completed successfully.
PS C:\Users\HP\Downloads>
```

```
commit 42967dffa9c1c3b73c657281c65be4442b277fd7
Author: [redacted] <[redacted]@gmail.com>
Date:   Tue Mar 24 20:33:00 2026 +0530
```

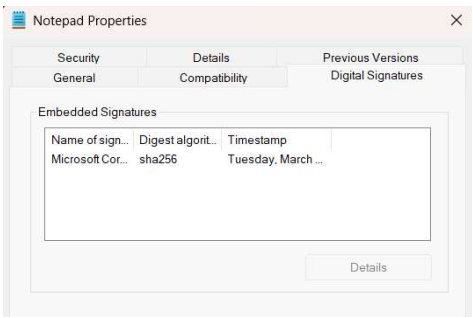
---

5. Digital Signatures and Certificates
SHA is used to generate message digests before signing.

- Ensures authenticity and integrity
- Used in government and enterprise systems

Examples:

- Digital signatures in PDFs
- Code signing in Microsoft and Apple ecosystems



---

6. Emerging / Alternative Use (SHA-3)
SHA-3 is not yet as widely used as SHA-2 but is gaining adoption.

- Used in high-security and future-proof systems
- Suitable for hardware implementations

Examples:

- Some IoT security systems
- Cryptographic research and next-gen protocols

In summary, SHA algorithms are deeply integrated into modern digital infrastructure—from securing websites and transactions to verifying files and enabling blockchain systems. Different versions are used depending on the required level of security and application needs.